# OVERALL REFINERY BLOCK FLOW DIAGRAM (BFD) & PROCESS INTEGRATION

BASIS: 100,000 BPD (564,670 kg/h) ARABIAN LIGHT / BRENT CRUDE BLEND | HIGH-CONVERSION CONFIGURATION

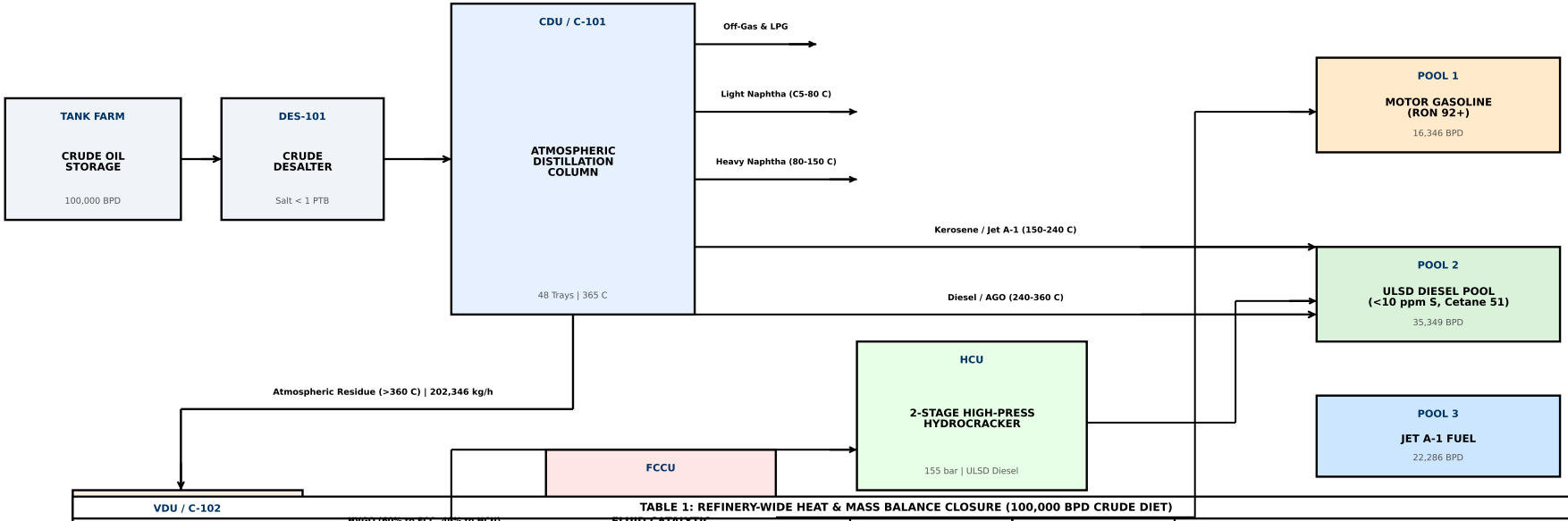


**TABLE 1: REFINERY-WIDE HEAT & MASS BALANCE CLOSURE (100,000 BPD CRUDE DIET)**

| STREAM / PRODUCT | FLOW (BPD) | MASS (kg/h) | WT % CRUDE | API GRAVITY | DOWNSTREAM DESTINATION |
|---|---|---|---|---|---|
| Crude Oil Feed (Arabian Light / Brent) | **100,000 BPD** | 564,670 kg/h | 100.00 % | 34.5 API | **Battery Limits -> Desalter -> CDU** |
| Blended Motor Gasoline (RON 92+) | 16,346 BPD | 80,670 kg/h | 14.29 % | 58.5 API | Gasoline Blending -> Commercial Pool |
| Ultra-Low Sulfur Diesel (ULSD, S<10ppm) | 35,349 BPD | 196,701 kg/h | 34.84 % | 37.8 API | Hydrotreated Diesel Pool -> Transport |
| Aviation Jet Fuel (Jet A-1 / Freeze -47C) | 22,286 BPD | 118,846 kg/h | 21.05 % | 42.8 API | Aviation Fuel Storage & Export |
| Liquefied Petroleum Gas (LPG C3/C4) | 16,435 BPD | 59,881 kg/h | 10.60 % | 105.0 API | Pressurized Horton Spheres / Commercial |
| Heavy Fuel Oil / Marine Bunker (380 cSt) | 2,587 BPD | 16,453 kg/h | 2.91 % | 10.5 API | Marine Fuel Oil Blending |
| Solid Petroleum Green Coke (Anode Grade) | 587 T/day | 24,461 kg/h | 4.33 % | 0.0 API | Hydraulic Decoking Pit -> Aluminum Smelter |
| Hydrogen Consumption (SMR Plant) | - | + 1,440 kg/h | + 0.25 % | - | Hydrocracking & Hydrotreating Saturation |
| Refinery Fuel Gas & Flare Loss | - | 42,670 kg/h | 7.56 % | - | Captive Fired Heaters & Safety Flare (API 521) |
| **TOTAL MASS BALANCE CLOSURE** | **100,000 BPD** | 564,670 kg/h | 100.00 % | - | MASS IN = MASS OUT (ZERO ACCUMULATION) |

**JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS PLANT DIGITAL TWIN SIMULATION**

| DRAWING TITLE | LAB / SUBJECT | COURSE / BRANCH | SCALE |
|---|---|---|---|
| REFINERY BLOCK FLOW DIAGRAM & MASS BALANCE | PROCESS PLANT DIGITAL TWIN & CAD | B.E. CHEMICAL ENGINEERING | DIAGRAMMATIC |
| **STUDENT NAME** | **ROLL NO.** | **DRAWING NO.** | **DATE** |
| SOURISH SENAPATI | 002310301035 | JU/ChE/VP/DWG-01 | 19 / 10 / 2026 |

VIRTUAL PETROLEUM REFINERY | 100,000 BPD CRUDE DIET | GAZEBO & ROS 2 DIGITAL TWIN | API / ASME BPVC